End-Tidal Carbon Dioxide ($EtCO_2$)

End-tidal carbon dioxide ($EtCO_2$) is the measurement of the concentration or partial pressure of carbon dioxide (CO_2) released at the end of exhalation. It is typically monitored using a technique called *capnography*, which provides both a numerical reading and a waveform that reflects ventilation status. Normal $EtCO_2$ values generally range between **35–45 mmHg** in healthy adults. Values below this range may indicate hyperventilation, poor cardiac output, shock, or pulmonary embolism, while elevated levels may suggest hypoventilation, airway obstruction, or respiratory depression.

In triage and emergency medicine, $EtCO_2$ is a highly valuable physiological marker because it provides rapid insight into both respiratory and circulatory function. Low $EtCO_2$ readings can occur when blood flow to the lungs is reduced, such as during severe shock or cardiac arrest. Conversely, elevated $EtCO_2$ may indicate inadequate ventilation or respiratory failure. Since $EtCO_2$ changes can occur earlier than oxygen saturation changes, capnography is often used for the early detection of patient deterioration.

$EtCO_2$ monitoring is especially important in critically ill patients, trauma cases, and during resuscitation. During cardiopulmonary resuscitation (CPR), $EtCO_2$ levels help clinicians evaluate the effectiveness of chest compressions and may indicate the return of spontaneous circulation (ROSC). It is also commonly used to confirm proper placement of endotracheal tubes after intubation.

Including $EtCO_2$ in triage datasets could improve the identification of patients at risk of respiratory or cardiovascular collapse. Because it provides continuous, non-invasive monitoring, it can support faster decision-making and improve patient prioritization in emergency settings.

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References

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